Binary Search Tree (BST)

write small

It is a special kind of BT where each node follows the BST property

⇒ Left Subtree of a node contain less than node's value

⇒ Right Subtree of a node contain values greater than the node's value.

The structure allows for efficient searching, insertion & deletion.

Inspired by Binary Search algorithm.

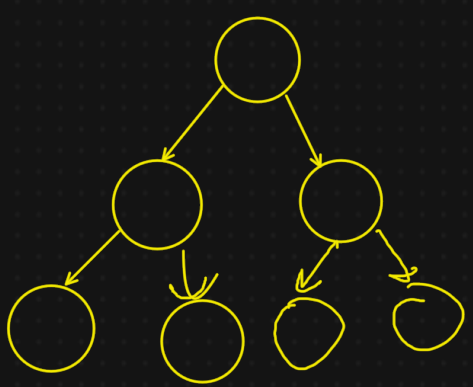
$O(n) \longrightarrow O(\log n)$ ⇒ if balanced.

Real world example

→ Phone book
→ dictionary
→ Auto complete feature (BST in background)
→ File searching

Industry

→ Databases (Indexing)
→ ML (Decision Tree)
→ Range Queries in DBs (amazon example)





Defining a BinarySTree / Point

Going to be same only.